A Project on

COMPARISION OF HYDROGEN INJECTORS

For the course of Case Study 1 Mech 6961 of Master of Engineering in Aerospace Engineering

Prepared For

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ABSTRACT 5
INTRODUCTION 6
LITERATURE REVIEW AND DESIGN BASIS 7
BOUNDARY CONDITIONS 8
4.1 Derivation of Inlet Conditions 8
4.2 Pressure 8
4.3 Velocities of Injection 8
4.4 Equivalence Ratio 8
4.5 Area Calculations 8
INJECTOR CONCEPTS 10
5.1 Concept 1 10
5.2 Concept 2 10
CFD METHODOLOGY 11
6.1 Geometry of Model 1 11
6.2 Geometry of Model 2 12
6.3 Mesh 12
6.4 Simulation 13
6.5 Limitations in Simulation 14
NON-REACTING FLOW ANALYSIS 16
7.1 Velocity 16
7.1.1 Inside the injector 16
7.1.2 Injector Exit Plane 16
7.1.3 In the Combustion Chamber 17
7.2 Temperature 17
7.2.1 Inside the injector 17
7.2.2 Injector Exit Plane 18
7.2.3 In the Combustion Chamber 18
7.3 H2 Mass Fraction 18
7.3.1 Inside the Injector 18
7.3.2 Injector Exit Plane 19
7.3.3 In the Combustion Chamber 19

7.4 Conclusion 20
REACTING FLOW ANALYSIS 21
8.1 Flame Behavior 21
8.1.1 On the Longitudinal Plane 21
8.1.2 Inside the Injector 21
8.1.3 Injector Exit Plane 22
8.1.4 Interpretation 22
8.2 Temperature Field 23
8.2.1 On the Longitudinal Plane 23
8.2.2 Inside the Injector 23
8.2.3 Injector Exit Plane 24
8.2.4 Inside the Combustion Chamber 24
8.2.5 Interpretation 25
8.3 Velocity Field 25
8.3.1 On the Longitudinal Plane 25
8.3.2 Inside the Injector 26
8.3.3 Injector Exit Plane 26
8.3.4 Inside the Combustion Chamber 27
8.3.5 Interpretation 27
8.4 H2O Mass Fraction 27
8.4.1 Inside the Injector 27
8.4.2 Injector Exit Plane 28
8.4.3 Inside the Combustion Chamber 28
8.4.4 Interpretation 29
8.5 Pressure 29
8.6 Temperature Across the Model 30
8.7 H2 Concentration Across the Model 31
8.8 H2OConcentration Across the Model 31
NOx 33
9.1 NOx Tendency 33
9.2 Flame and Temperature Structure 33

9.3 Caution on Interpretation 34
9.3.1 Turbulence Chemistry Interaction Model 34
9.3.2 Grid Resolution Sensitivity 34
9.3.3 NOx Production 34
RESULTS AND DISCUSSION 36
10.1 Engineering Comparison 36
10.2 Compromises 37
10.3 Recommended Concept 37
10.4 Future Work 38
CONCLUSION 39
11.1 Technical Conclusions 39
11.2 Knowledge Gained 39
APPENDIX 40

ABSTRACT

Hydrogen is considered to be the cleanest and most promising fuel for the next generation of aircraft engines. Aviation accounts for approximately 3% of global CO_2 emissions, and hydrogen fuel has emerged as a good alternative to conventional fuel. Hydrogen injectors are critical in enabling technology for hydrogen powered aircrafts as they are responsible for precise metering, atomization and delivery of hydrogen into the combustion chamber under extreme pressure and temperature. Unlike conventional fuel, these hydrogen injectors must handle the fuel with different properties since its density is very low, the wide flammability limits of hydrogen and high diffusivity. This makes the design a very complex and critical challenge for safety.

Significant advancements have been made in the injector geometry optimization including multipoint and coaxial injector configurations that improve fuel-air mixing, making it uniform, and reducing the formation of NO_x value at high thrust and temperature. The development of cryogenic hydrogen injector systems which are capable of handling liquid hydrogen (LH_2) at very low temperature has been accelerated through Airbus's ZEROe initiative and various other research including the NASA's experimental hydrogen propulsion research. Improvement of Computational Fluid Dynamics (CFD) tools has also led to better conditions to simulate injector spray patterns, combustion stability, and flame holding behavior.

Hydrogen injectors when mature will be the primary component of the net-zero aviation, which the aviation industry is leaning towards, enabling the turbofan and turbo prop engines to operate with water vapor as the primary exhaust product rather than the CO_2 we have now. As the entire industry adopts cleaner fuel, the cost of production of these engines will also reduce, thus promoting the future of aviation.

INTRODUCTION

The primary aim of this project is to design 2 different Hydrogen fuel injector for an aerospace engine and simulate their working conditions in ANSYS Fluent under species transport and compare them. This project report will go in detail about the designs used, combustion model, assumptions and specifications, flame location and the shape of the flame, the temperature field at various planes in the model, regions of peak heat release and the reason behind it and finally the comparison between our models which includes key parameters such as NO_x and mixing of the fuel-air mixture.

For our project, we chose to go with a few control parameters as they set a baseline on where we can expect the changes to stem from. These parameters are the fuel to air ratio, and by having the same fuel to air ratio, we will have the same fuel and air inlet area in both our models. The velocity and pressure of injection are also taken to be constant across both models. The number of fuel holes changes from one model to another to understand which one has better results. The fuel we took is Hydrogen and the oxidizer is air with 23.3% oxygen and the rest being nitrogen. Since we took the fuel to air ratio to be a control, we got the mass flow rate of fuel to be 6.225 Kg/s and the mass flow rate of hydrogen to be 0.1 Kg/s in both cases.

LITERATURE REVIEW AND DESIGN BASIS

Lately, engineers have made real progress with hydrogen fuel gas turbines, especially when it comes to how you mix fuel and air. Getting the injector and combustor design right matters a lot—otherwise, you end up dealing with flame flashback, unstable combustion, or high NOx emissions. Hydrogen's kind of tricky because it burns fast and ignites easily, so you really need to be careful with how you inject it.

One of the best ideas to surface recently is combining swirl-stabilized burners with micromix and multi-point injection. Basically, when you create enough turbulence and shear flow, hydrogen and air blend together much quicker. The setup injects hydrogen radially and shoots air axially—using concentric pipes—so the two streams meet at right angles. That's way better for mixing than just lining them up.

To keep flashback away, tiny swirlers come into play—right in the premixing region. This whole system boosts combustion efficiency and lets you run lean, but it comes with a cost: you lose some aerodynamic efficiency.

As for the nozzle design, here's the plan—stick a swirler in to spin the flow like a vortex. Hydrogen blasts in sideways from several little holes, while air moves straight in through channels packed around the core. You get serious turbulence inside, thanks to the way high-speed hydrogen jets hit the slower-moving air. It's a compact setup that lowers flashback risk, but honestly, you still have to tackle the pressure loss problem.

Recent research backs up this approach. If you want peak performance, you need to inject hydrogen from multiple points radially into the chamber, while air pours in axially. Micromix technology uses nozzles that shoot hydrogen out fast, where it clashes with the incoming air in a tight space. The high-speed jets create a shear layer, which stirs up turbulence and really mixes the gases. Plus, this technique does two important things for hydrogen combustion: it cuts down NOx emissions and helps keep flashbacks under control.

In practice, the device sends air through two concentric channels, with hydrogen jets entering from multiple holes around the injector's face. The hydrogen shoots right into the airflow, so you get a turbulent zone inside the combustor. The main drawback? Using lots of injection points tends to increase pressure loss, which is something you can't ignore.

BOUNDARY CONDITIONS

The boundary conditions for the models were chosen with a few of the basic parameters same across both the models and the rest were calculated based on the constant parameters. For instance, the amount of fuel holes changes between the models but the total area of fuel injection was taken to be constant, along with the velocity and the pressure of injection. We will look at the parameters a bit more in detail.

4.1 Derivation of Inlet Conditions

The inlet conditions as mentioned earlier were calculated from set parameters. Those parameters were,

4.2 Pressure

Pressure was chosen to be constant at 15 bar (1500000 Pa). This was the pressure of injection and the pressure at which the entire combustion chamber functions. It was chosen as it represents a standard operating midpoint for many modern aircraft gas turbine engines.

4.3 Velocities of Injection

The Air was chosen to be injected at 100 m/s at 700K, and the fuel was chosen to be injected at 600 m/s at 300k.

4.4 Equivalence Ratio

The equivalence ratio was chosen to be 0.55 based on the literature review and is common in hydrogen injectors. According to this equivalence ratio, the air to fuel ratio was calculated to be 62.25. Based on this, the fuel mass flow rate was to be 0.1 kg/s, and the air mass flow rate was 6.225 kg/s. These mass flow rates were constant across the designs used.

4.5 Area Calculations

Based on all the values from before, when substituted in the equation,

$$\dot{m} = \rho \times A \times V$$

Where the mass flow rates are, 6.225 kg/s for air and 0.1 kg/s for fuel, velocities are 100 m/s and 600 m/s respectively. The densities are 1.21 kg/m^3 for fuel and 7.47 kg/m^3 for air, we get the total areas of injection to be,

$A_f = 0.00013774\ m^2$

$A_a = 0.0083333\ m^2$

And the area of fuel was divided into multiple smaller regions spread across the inner and outer walls of the injector to produce better mixing.

INJECTOR CONCEPTS

As mentioned earlier, the design of the two different models was based on set parameters with same mass flow rates for the inlets. But the concepts will be looked at in this chapter.

5.1 Concept 1

The design of concept 1 was inspired by swirl injectors that induce rotating, vortex motion into propellants, especially in the hydrogen air mixture that we are designing it for. Since the fuel (Hydrogen) is injected at a higher velocity, it was injected radially to the entire flow of the propellants and the injector, whereas the air was injected axially to the injector as it was at a lower velocity. This helps us in the mixing of the hydrogen and air mixture better than introducing them both in the same direction. The swirlers were intentionally designed a bit smaller so as to prevent flashbacks in the swirler section. The mixing of the fuel and air mixture is considered to be its strength, whereas the loss of pressure might be a drawback to this design.

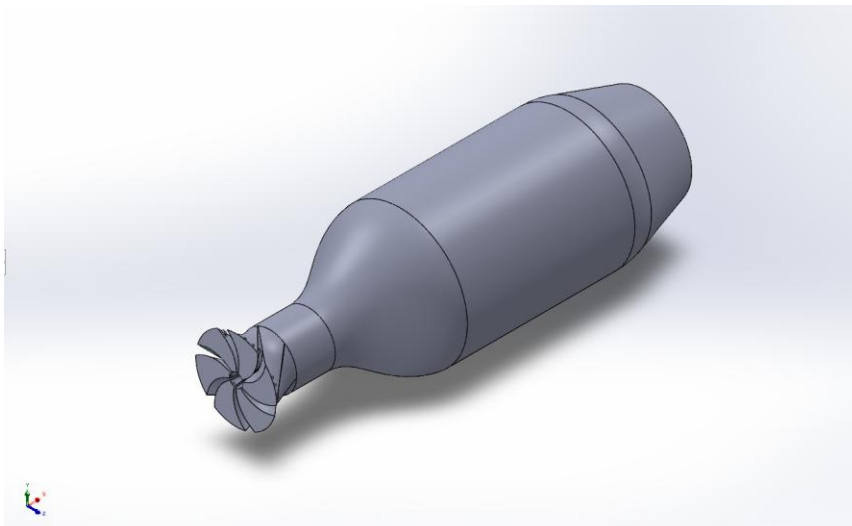
5.2 Concept 2

The design concept of the injector 2 is based on the axial introduction of air through the two concentric circular passages. Hydrogen fuel is supplied through multiple small orifices distributed on the cylindrical face and the circular air inlet face. These orifices allow hydrogen to be injected radially at high velocity into the surrounding airflow. As a result, the fuel enters the system from several directions—moving inward from the outer annular region and outward from the central section—thereby penetrating directly into the main air stream. The use of multiple radial injection points, together with the slower axial airflow, creates pronounced shear layers at the injector outlet. These interactions lead to increased turbulence, which in turn improves the mixing of hydrogen with the air.

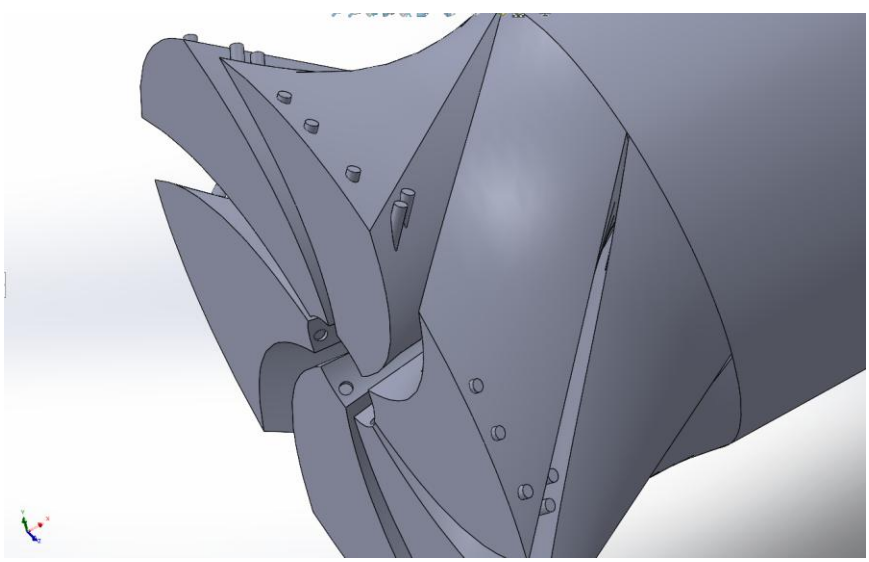
CFD METHODOLOGY

6.1 Geometry of Model 1

The geometry of model 1 as mentioned in the previous chapter will include a swirler which helps in the mixing of fuel and air, the rest of the injectors is 150mm which expands into a chamber section and terminating in a nozzle section at the end where the NO_x value will be recorded. The fuel injectors are radial to the overall flow while the air is injected axially. A total of 36 holes is present for the injection of fuel, each hole having an area of 0.00001147 m^2.



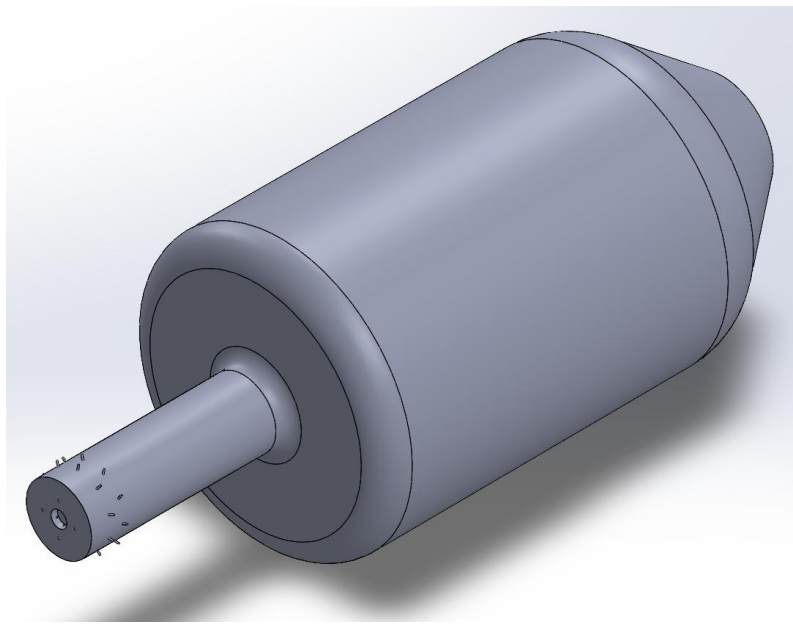
Geometry of Model 1



Fuel inlet of Model 1

6.2 Geometry of Model 2

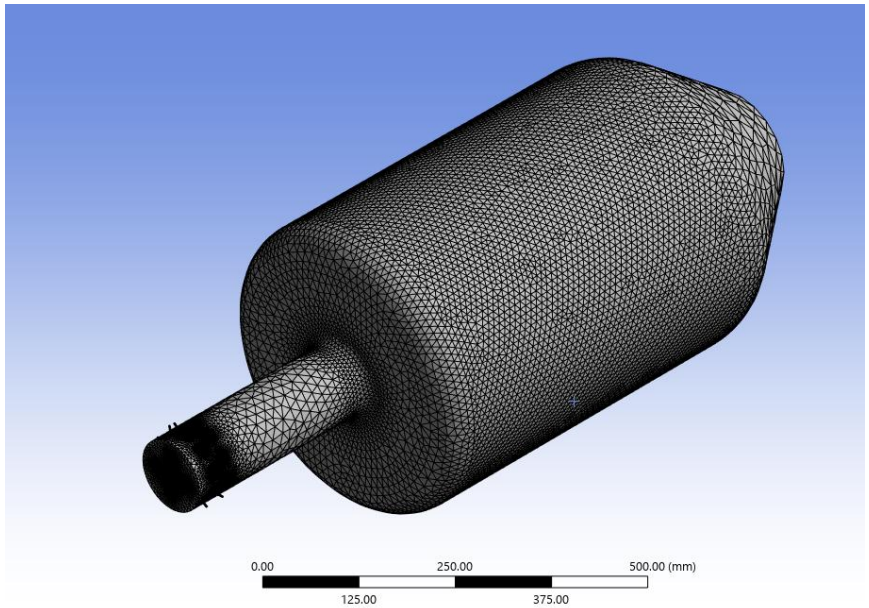
The geometry model 2 has axial air introduction and radial fuel inlets as mentioned above. A total of 31 fuel holes is present in the injector. Out of these, 26 holes have an individual area of 3.84×10^{-6} m², while the remaining 5 holes each have an area of 7.65×10^{-6} m². Fuel is injected radially through these holes, while air enters axially to facilitate proper mixing before proceeding to the chamber and nozzle section.



Geometry of Model 2

6.3 Mesh

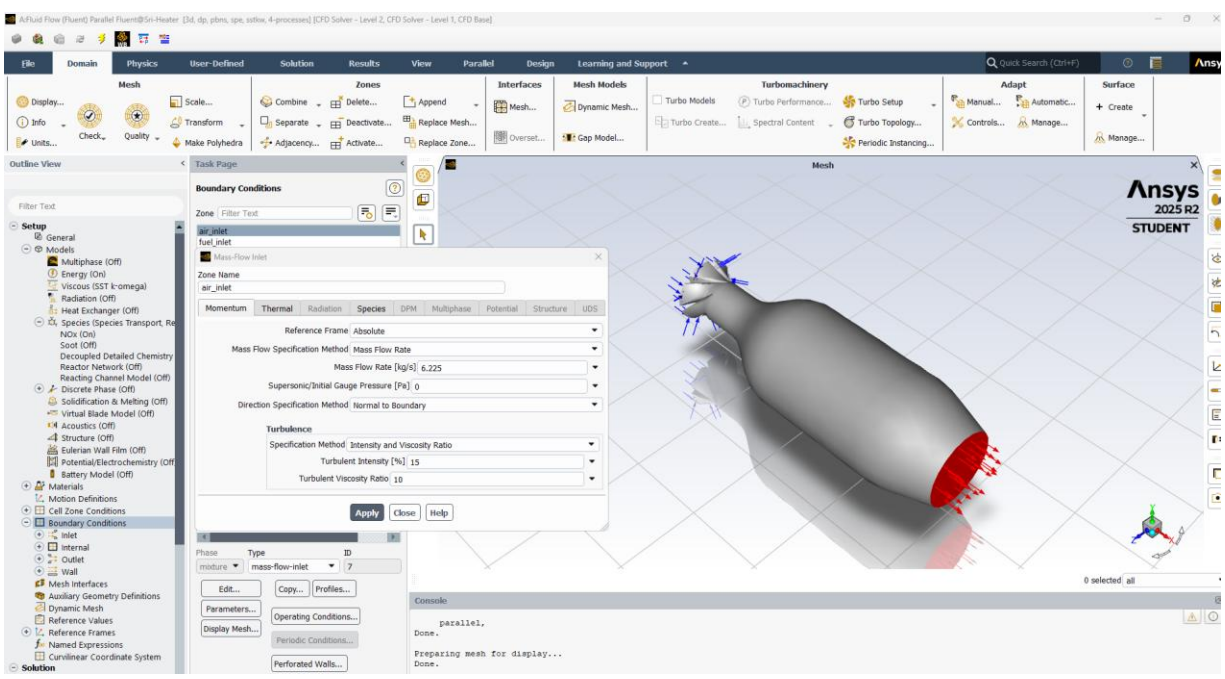
A coarse mesh was done with a similar and very high number of elements with the main focus on the quality of the mesh for both the models.



Mesh of Model 2

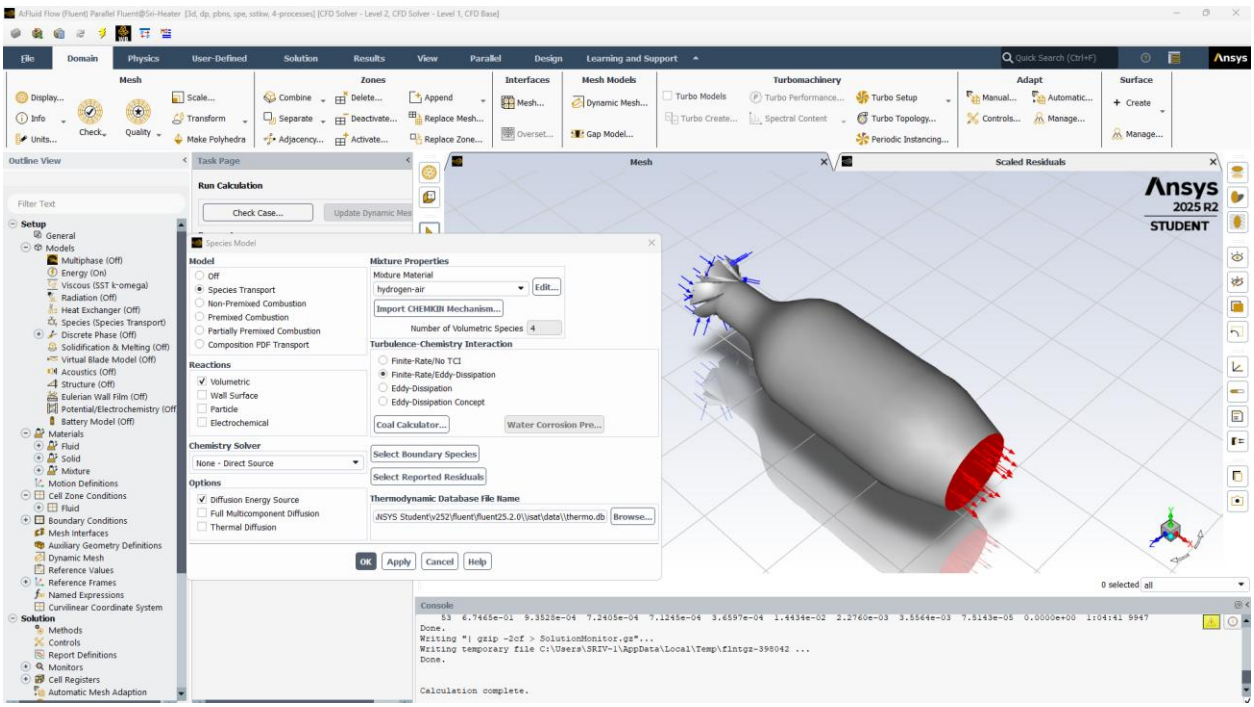
6.4 Simulation

The boundary conditions given were based on the parameters calculated. The inlet was selected to be a mass flow inlet with the inlet air mass flow of 6.225 kg/s and the temperature to be 700 K and the inlet fuel mass flow to be 0.1 kg/s with the temperature to be 300 K. The mass fraction of the fuel inlet was 100% hydrogen with the mass fraction of the air inlet was 23.3% oxygen and the rest was nitrogen.



Fluent Setup with Air Inlet of Model 1

The combustion model used was species transport with the mixing material as hydrogen-air. The turbulence chemistry used was Finite-Rate/Eddy-Dissipation. For the cold flow simulations, the model was simulated without Reactions turned on. But for the final combustion model analysis, the reaction was turned on with volumetric setting. The simulation was run for a preliminary 3000 iterations, thus concluding non reacting flow simulations. For reacting flow simulations, after preliminary iterations, a patch region was made in the shape of a sphere inside the combustion chamber. The temperature of 2000 K and H_2O as 0.01 was patched into this region to simulate combustion. NO_x was turned on simulated for 2000 more iterations.



Combustion Model used in Model 1

After the models were simulated, the solutions section was used to obtain various contours such as velocity magnitude, heat of reaction, temperature, mass fraction of species, etc., at various planes and surfaces of the simulated model. It was also used to obtain the mole fraction of NO_x at the outlet which was used to calculate the NO_x parts per million values.

6.5 Limitations in Simulation

The results we obtain can be prone to limitations due to the turbulence which can be found in real-world conditions, and it may not be simulated as is, in Ansys Fluent. Turbulence plays an important role in the mixing of the fuel and air in the injector and can lead to varying results if not simulated accurately. The other limitation can be expected in the meshing front, as the number of elements, even though high, is relatively low when

compared to industry standards, which exceeds 10 million elements. This can prove to be one of the major reasons for not being able to achieve convergence.

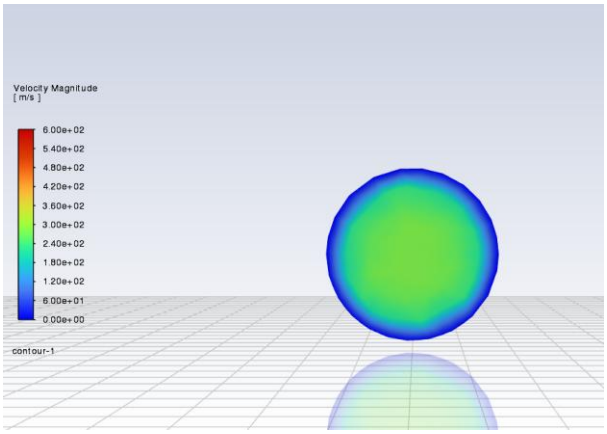
NON-REACTING FLOW ANALYSIS

Both the models were simulated first for non-reacting analysis to understand the mixing of the fuel and air.

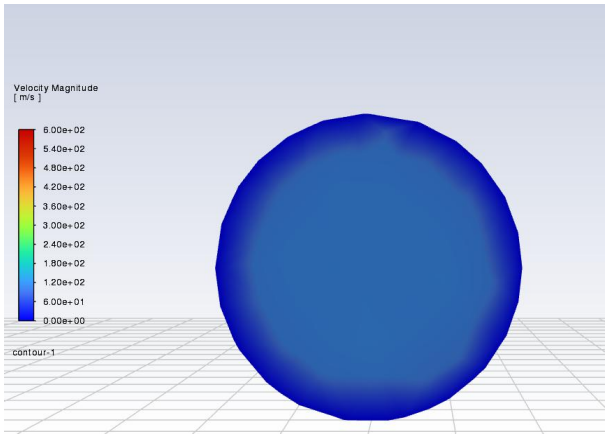
7.1 Velocity

7.1.1 Inside the injector

Model 1 shows a well-structured annular velocity profile. It has a green core which denotes velocities between 120 m/s and 180 m/s at the injector mixing plane and a blue wall boundary layer, thus indicating a swirl driven jet in action. Whereas model 2 at the same zone shows an almost uniform blue field which indicates 0 to 60 m/s, which indicates that the flow is slow and structureless with the velocity spiking only at the injection of the fuel (hydrogen). Thus, indicating superior velocity structure in model 1.



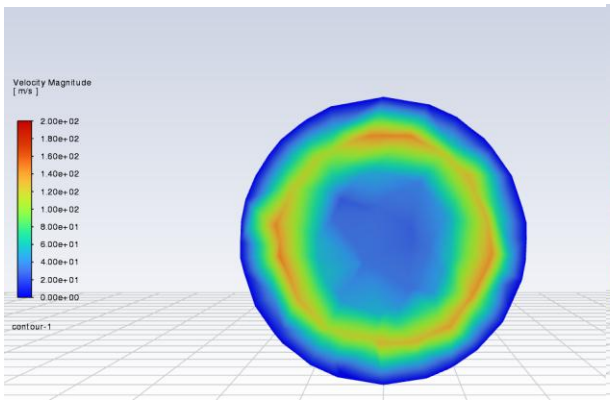
Velocity Magnitude of Model 1



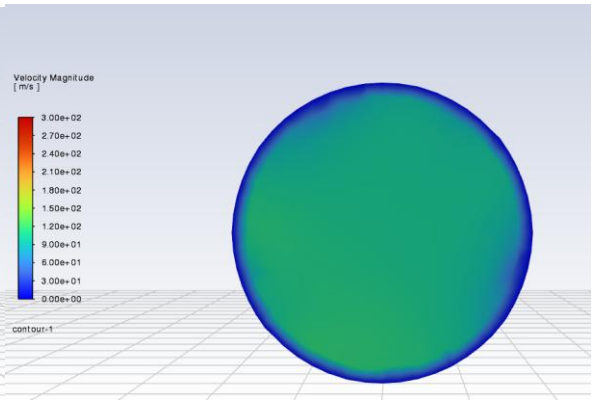
Velocity Magnitude of Model 2

7.1.2 Injector Exit Plane

Model 1 shows an annular high velocity ring at the exit of the injector (120 m/s to 160 m/s) with a low velocity core. This is the signature of central recirculation zone formation. This is an important feature for the flame to stabilize in a gas turbine injector. Whereas model 2 shows a uniform profile which ranges from 120 m/s to 150 m/s with a thin boundary layer and no formation of central recirculation zone. The CRZ gives model 1 a major advantage as it will recirculate the hot combustion products back towards the injector face in a reactive simulation, thus has a higher potential to be the better model out of the two.



Velocity Magnitude of Model 1



Velocity Magnitude of Model 2

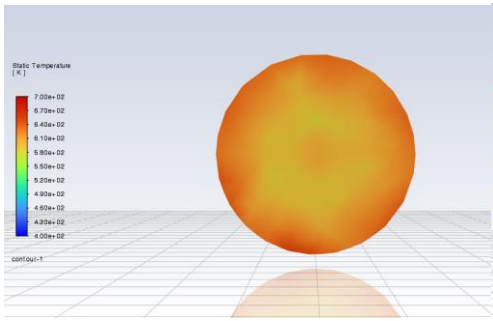
7.1.3 In the Combustion Chamber

The model 1 has a donut shaped velocity distribution while compared to the model 2's almost entirely deep blue field with only faint off center smear of green regions. This chamber is significantly stagnant than compared to the model 1 with higher velocity. We can also find the effect of the induced swirl during the start of the injector carrying forward all along the model. Model 2's low chamber velocity means it has low turbulence, poor convective heat transfer and very slow reaction rates when combustion is enabled. Thus Model 1 is the better prospect when it comes to the combustion chamber.

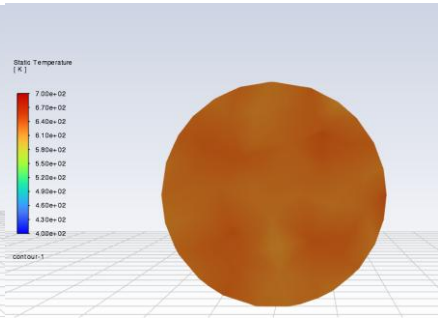
7.2 Temperature

7.2.1 Inside the injector

Model 1shows a uniform temperature region ranging between 650K and 700 K. The preheated air that is injected axially dominates and the radially injected fuel acts as a cooling effect for the entire flow and we can see that the fuel has been absorbed into the bulk faster than the mixture in model 2. But the model 2 accounts for it with a longer injector design, the important thing to note here is that both the models avoid any cold pockets, so one model is not evidently better than the other based just on the temperature at the injector.



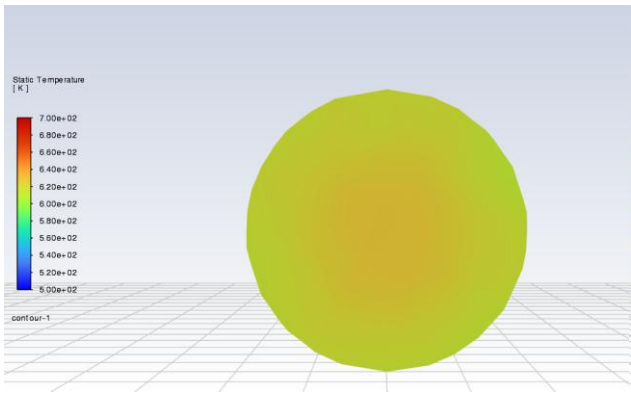
Static Temperature of Model 1



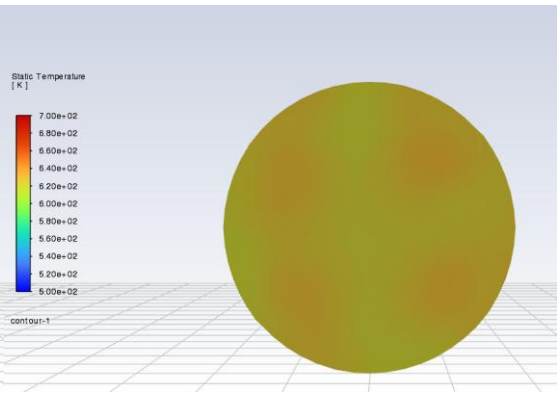
Static Temperature of Model 2

7.2.2 Injector Exit Plane

At the injector exit, model 1 shows a yellow to green shade denoting the temperature to range between 580K to 620K with visible cooler pockets. This indicates that there is still active mixing in that location. On the other hand, model 2 shows uniform yellow to greeen profile with almost no variation such as the one found in the other model. While it may look good at first glance, reflects passive equilibration rather than the turbulence driven mixing that we are looking for. This mixing can be found in the first model, and it shows ongoing mixing as well.



Static Temperature of Model 1



Static Temperature of Model 2

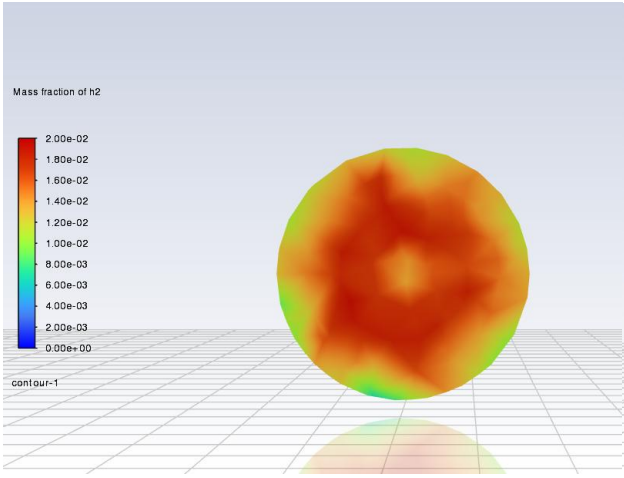
7.2.3 In the Combustion Chamber

Both models show remarkably similar thermal uniformity in the combustion chamber, as they both are in a small difference with each other (10K). While model 1 shows a difference of 10K, model 2 shows a difference of 8K or less. However, while looking at the temperature and the velocities at the same location, we can find that model 1 achieves similar result despite having higher velocity and thereby higher turbulence. But model 2 achieves this result while being almost stagnant.

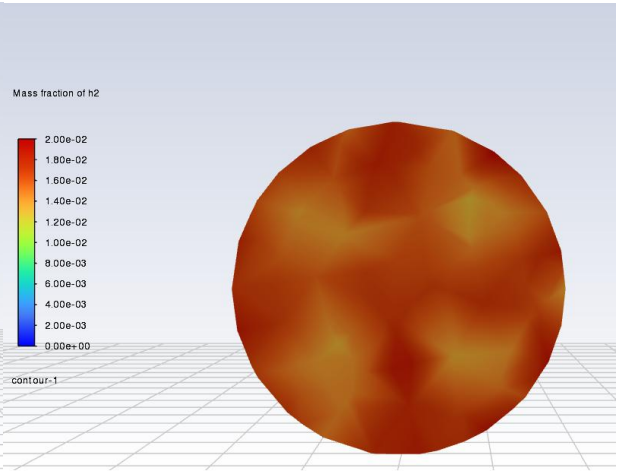
7.3 H_2 Mass Fraction

7.3.1 Inside the Injector

Model 1 shows a patchy swirl pattern as it is very close to the exit of the swirlers and the mixture is in rotatory motion, which does not mean uniform mixing but shows the mixture is energetically active with strong diffusion. Model 2 shows uniform distribution of H_2 across the surface which can be misleading as there is no swirl involved or any other method of mixing involved such as turbulent mixing. Instead, the long tube allows for passive diffusion. The mechanism in model 1 is more reliable and efficient than model 2.



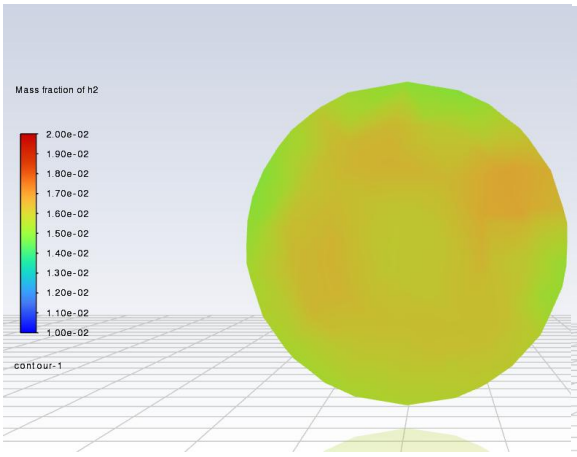
Mass fraction of H_2 of Model 1



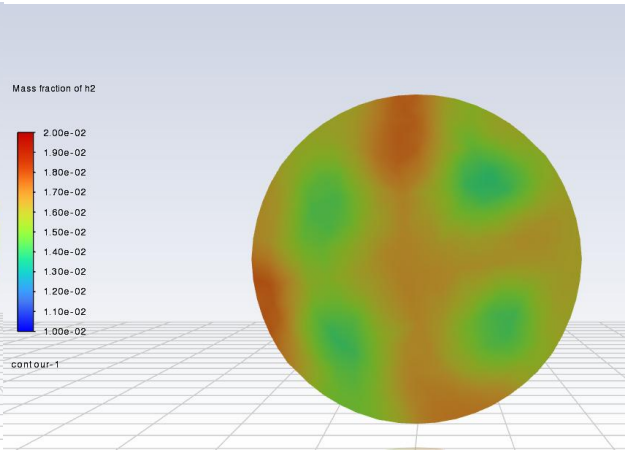
Mass fraction of H_2 of Model 2

7.3.2 Injector Exit Plane

At the exit of the injector, we can observe a small asymmetry in the distribution of model 1. This may be due to the swirl induced and not a serious concern since the downstream chamber area expands suddenly and corrects it quickly. But model 2 shows large scales of non-uniformity with large green and dark orange patches which are hard to correct. The mixing characteristics at the exit of the injector plane is better in model 1 despite its swirl induced asymmetry.



Mass fraction of H_2 of Model 1

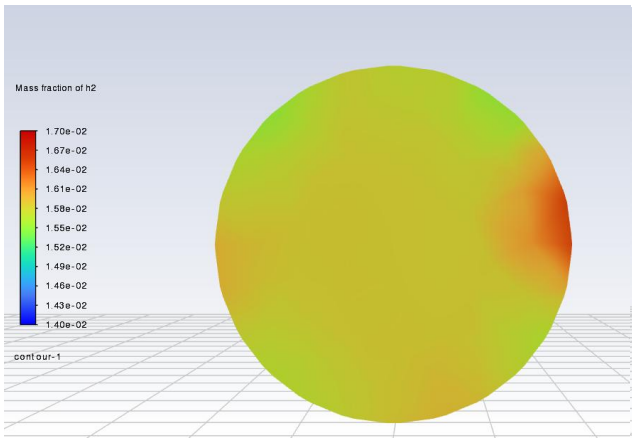


Mass fraction of H_2 of Model 2

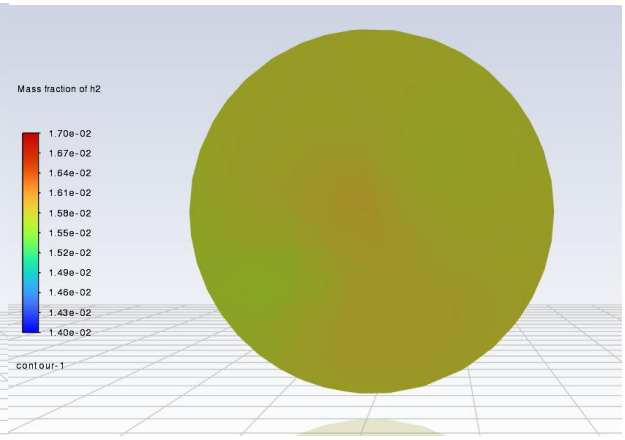
7.3.3 In the Combustion Chamber

This is first place where both the models are comparable. Model 1 shows a small richer pocket on one side. Model 2 shows nearly perfectly uniform yellow to green distribution, slightly more uniform than model 1. Both models have successfully homogenized the H_2 – air mixture by the time the flow reaches this point. Even though they have similar mixing,

the velocity and temperature conditions vary by a huge margin since the path they took to achieve it was different.



Mass fraction of H_2 of Model 1



Mass fraction of H_2 of Model 2

7.4 Conclusion

Model 1 wins at almost every meaningful location and parameter. The chamber level H_2 , temperature distribution, uniformity were are comparable but the path taken by the two models are different. Model 1 achieved it with swirl, having high turbulence and CRZ which made it possible in a short mixing length and made it more efficient, whereas model 2 achieved it by diffusivity and longer injector length, lower velocity, no recirculation, thereby increasing the residence time. It would be unable to sustain stable combustion. For a hydrogen fuel injector model 1 proves to be a clear winner and accounts for combustion intensity, flame stabilization and flashback resistance.

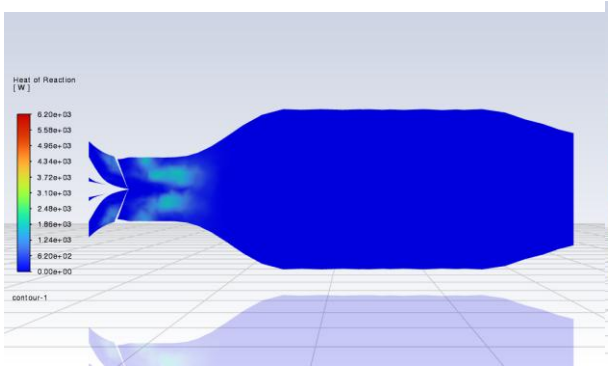
REACTING FLOW ANALYSIS

Both the models have the same combustion chamber geometry, a cylindrical injector tube feeding into a large combustion chamber. The key structural difference is the injector design configuration. Model 1 has a more compact injector design since it has swirlers to help it in mixing than the long injector design found in model 2 since it relies on diffusivity over a longer area.

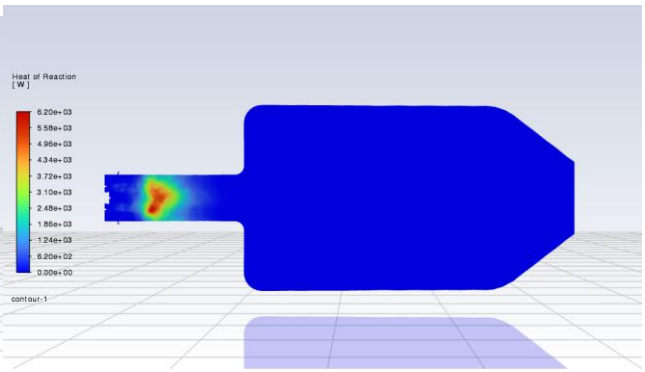
8.1 Flame Behavior

8.1.1 On the Longitudinal Plane

Model 1 has a combustion confined to the injector zone but the reaction zone diffuse and spread axially into the combustion chamber denoting it as green field and low intensity cyan color field. Model 2 also shows a similar pattern where the flame is also confined to the injector zone but it does not seem to diffuse and spread axially into the combustion chamber and is localized to the injector.



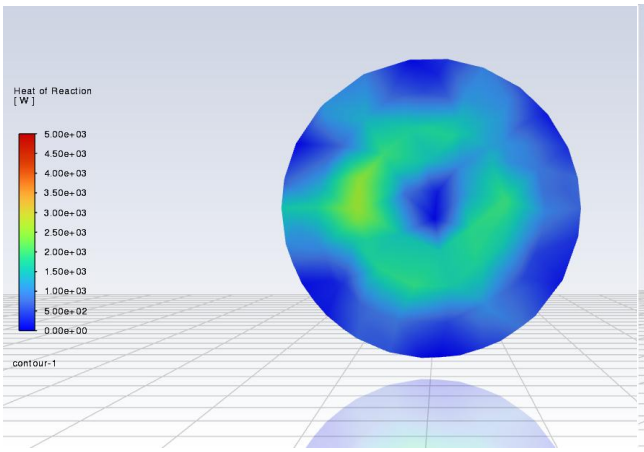
Heat of reaction of Model 1

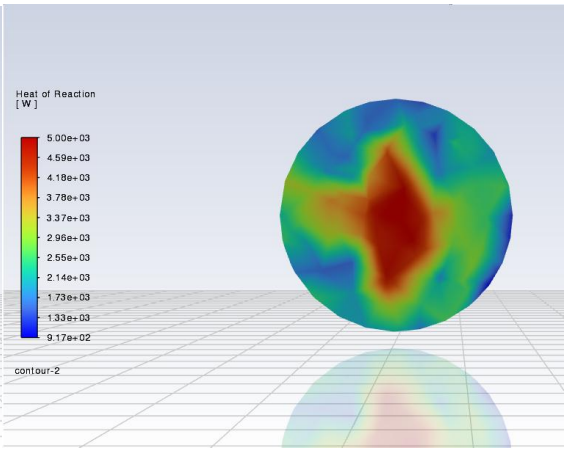


Heat of reaction of Model 2

8.1.2 Inside the Injector

Model 1 has reaction heat that is broadly distributed across the entire surface of the plane. This denotes that the flame is not localized. This is the characteristic of a well-mixed swirl flame, where the fuel and air mixture have already mixed before the reaction zone. On the other hand, model 2 shows a heat of reaction peak of 5000 to 6200 W which is concentrated in a compact and asymmetric hotspot. We can also observe that the flame is anchored close to the injector exit. The reaction zone is narrow and intense, thereby indicating the formation of a short, attached diffusion style flame.

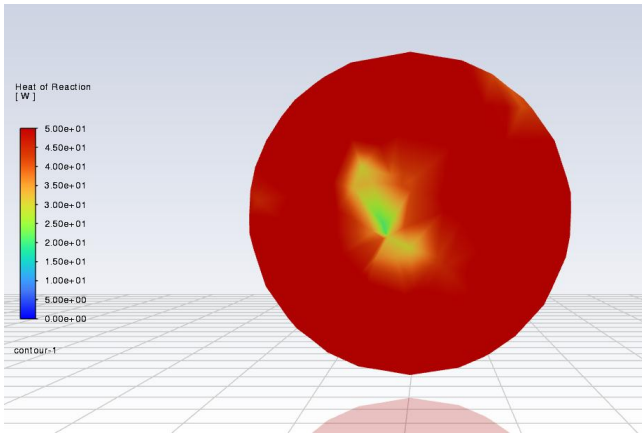


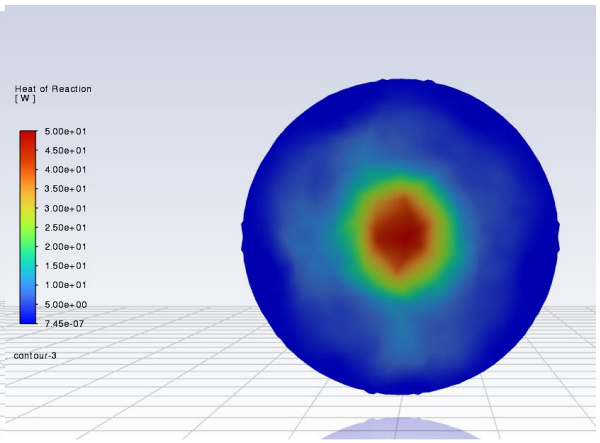


Heat of reaction of Model 1 Heat of reaction of Model 2

8.1.3 Injector Exit Plane

Model 2 shows a uniform red field covering the entire cross section denoting that the reaction is still occurring broadly across the entire plane. This means that the combustion has not completely ended and will continue into the combustion chamber. But in model 2, we can see a centrally localized compact hot spot. This means combustion is essentially complete by this point and the only remaining part is localized in the center.





Heat of reaction of Model 1 Heat of reaction of Model 2

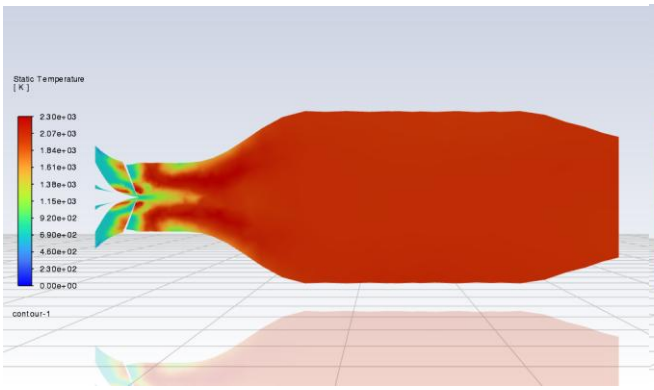
8.1.4 Interpretation

Model 1 produces a distributed, volumetric flame which burns more gradually over the entire region, whereas model 1 produces a more concentrated and fast burning flame which has high local heat release. Model 1 is more desirable as it has better mixing and for a uniform heat loading on the combustor walls.

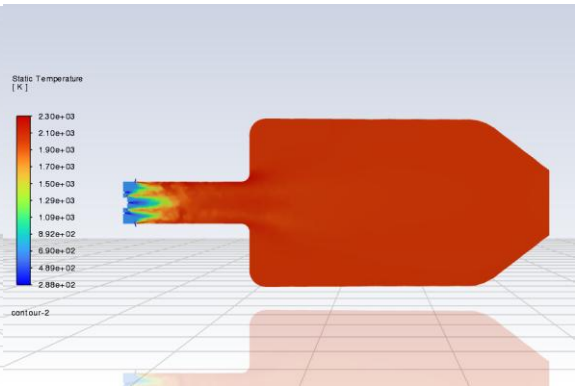
8.2 Temperature Field

8.2.1 On the Longitudinal Plane

Both models show the large combustion chamber fully saturated at 2300K confirming combustion and high enthalpy in the main chamber for both the models. The injector tube shows the cold to hot transition zone, the velocity inside the combustion chamber in model 1 looks more promising than the one found in model 2.



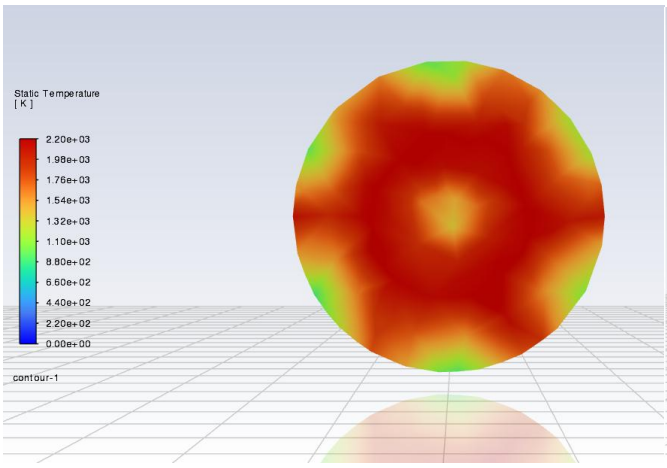
Model 1 Static Temperature



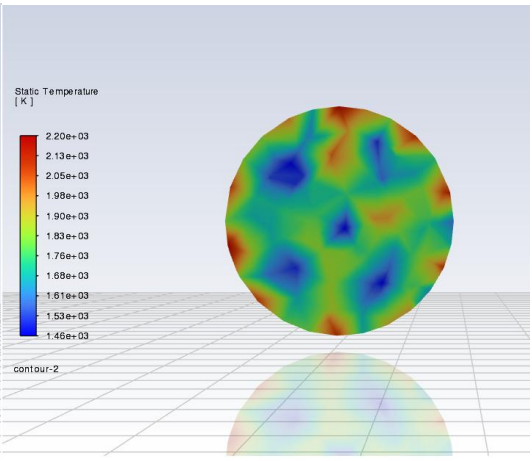
Model 2 Static Temperature

8.2.2 Inside the Injector

In this plane, a smooth, concentric annular temperature ranging between 1800K to 2300K at an intermediate radius is hot and cooler in the center and the outer wall. This is the hallmark of swirl stabilized flame with central recirculation zone (CRZ). The cooler center is the CRZ where burned products recirculate, flanked by the primary reaction annulus. Model 2 shows a highly non uniform and patchy temperature field ranging between 1400K and 2200K with irregular hot and cold patches scattered across the plane. This shows incomplete mixing with fuel rich and fuel lean zones.



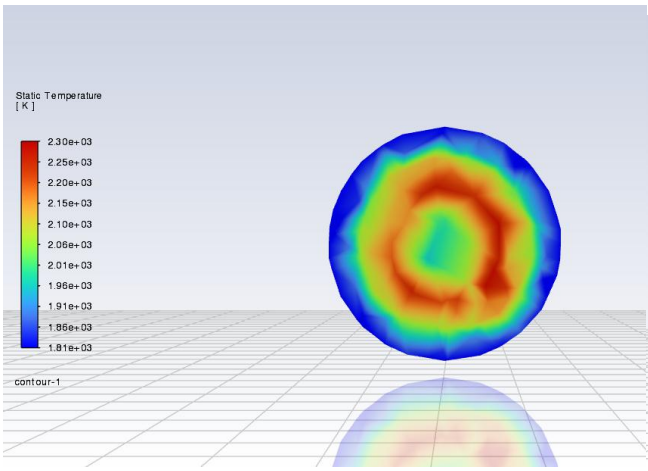
Model 1 Static Temperature



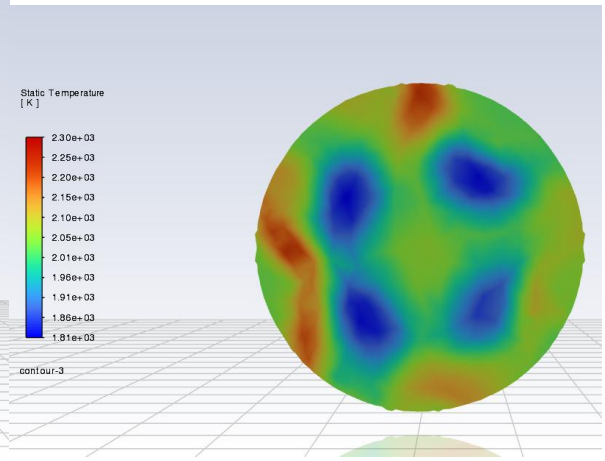
Model 2 Static Temperature

8.2.3 Injector Exit Plane

Model 1 shows a predominantly red and uniform temperature field between 1800K and 2200K only showing a mild variation across the plane section and is thermally more homogeneous, indicating better mixing and more complete combustion at this location. On the other hand, model 2 has a range of 1800K to 2300K still with a patchy and non-uniform distribution of hot and cold spots alternating with each other. These spots correspond to the positions of the individual injector holes, and the jets have not fully mixed yet.



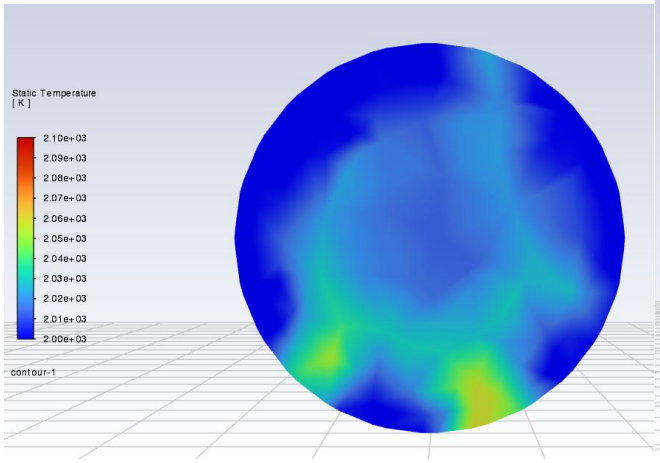
Model 1 Static Temperature



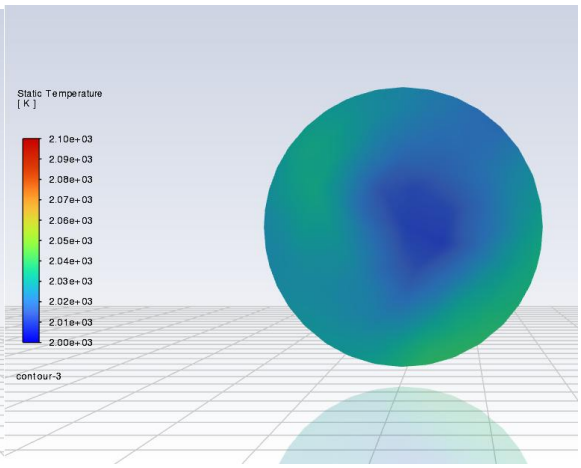
Model 2 Static Temperature

8.2.4 Inside the Combustion Chamber

Model 1 shows a more spatially spread zone with a possibility of a cool zone at the lower half of the cross section while the upper portion remains at 2050K to 2100K, thus indicating that residual mixing is ongoing at this location. On the other hand, model 2 shows a temperature range between 2000K to 2100K with a very smooth and nearly uniform field in the chamber. Even though it spreads very well, a slightly cooler zone can be found in the center reflecting full mixing and post flame equilibrium.



Model 1 Static Temperature



Model 2 Static Temperature

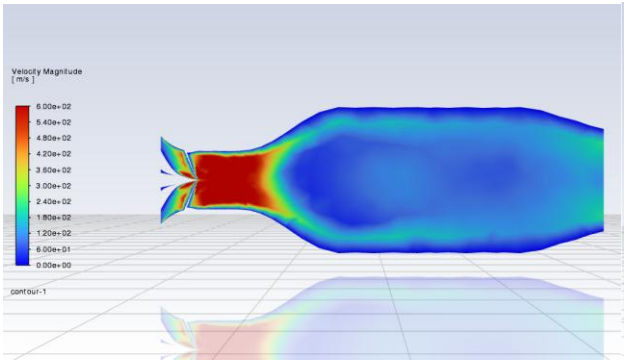
8.2.5 Interpretation

Model 1 shows a well-structured swirl flame in the injector but less uniformity far downstream, which suggests the length for mixing is longer, but on the other hand, model 2 shows excellent thermal uniformity deep in the chamber but has very patchy temperatures in the model, which may mean it its unreliable and has hot spots, which is not desired in a injector.

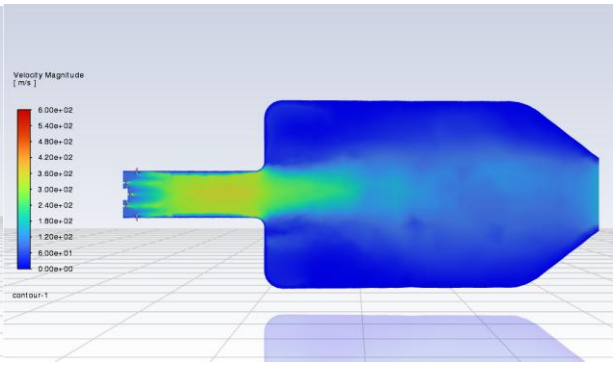
8.3 Velocity Field

8.3.1 On the Longitudinal Plane

The model 1shows a higher velocity in the xy plane at around 830m/s while model 2 showed a maximum of 600m/s. This can be due to the reduction in area from the injection of air and fuel in model 1. Whereas model 2 has a bigger radius and is longer than the other model to account for diffusivity for the mixing of fuel and air.



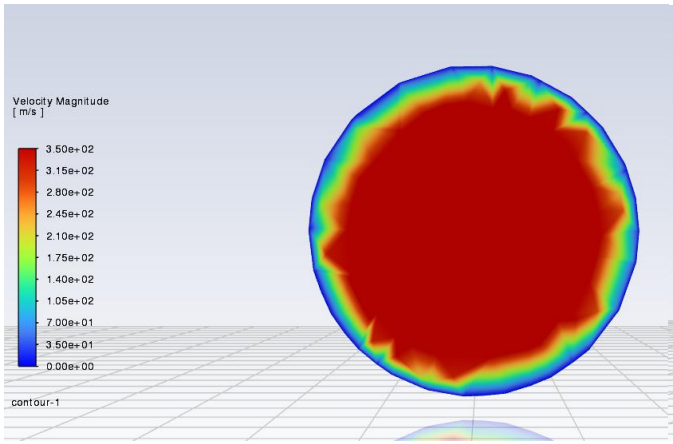
Model 1 Velocity Magnitude



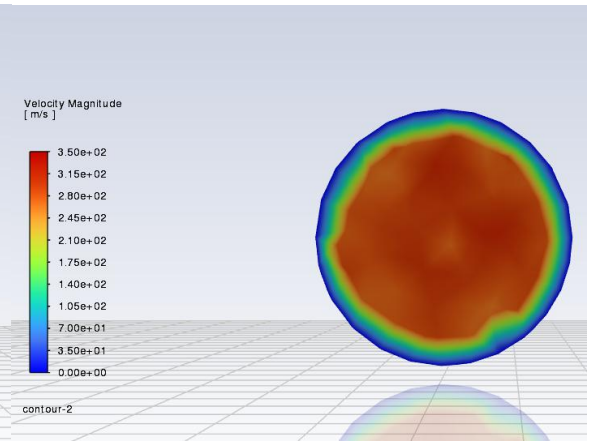
Model 2 Velocity Magnitude

8.3.2 Inside the Injector

Model 1 shows a maximum of 350 m/s but with a clear annular velocity profile with a velocity ring. The velocity at the center is low at around 90 m/s and near zero at the wall. This is the textbook swirl induced velocity distribution with a low velocity CRZ at the center. Model 2 shows a nearly solid red disk, meaning the injector is a high-speed plug flow and not mixing well.



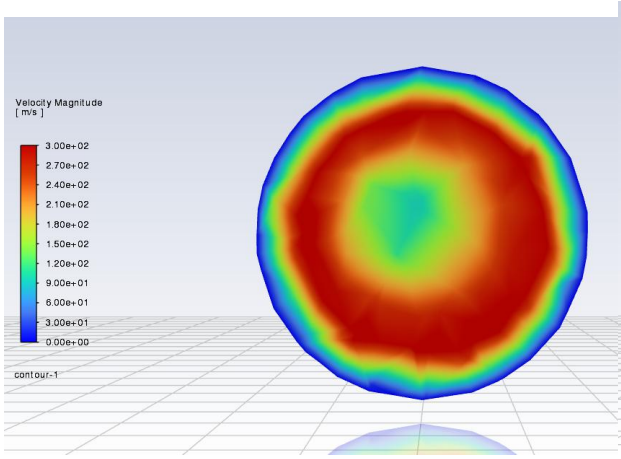
Model 1 Velocity Magnitude



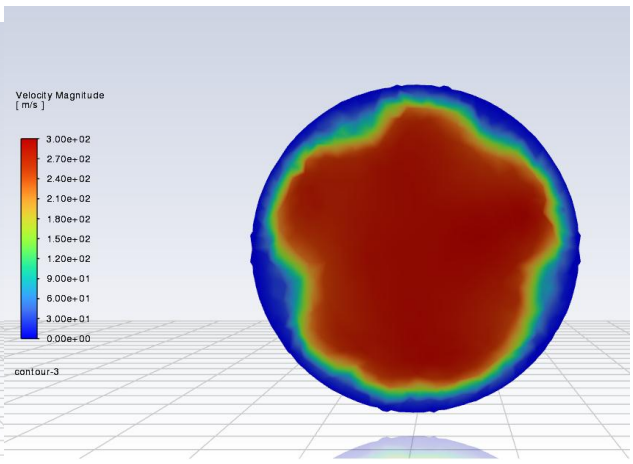
Model 2 Velocity Magnitude

8.3.3 Injector Exit Plane

The same phenomenon observed inside the injector can be observed at the end of the injector plane aswell. Model 1 shows a maximum of 300 m/s annular ring surrounding a lower velocity center which is reflected as a result of the swirl induced. Model 2 has the same plug as profile which was observed in the previous plane.



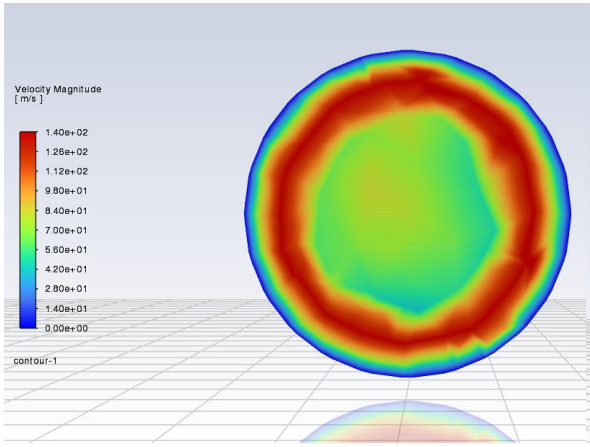
Model 1 Velocity Magnitude



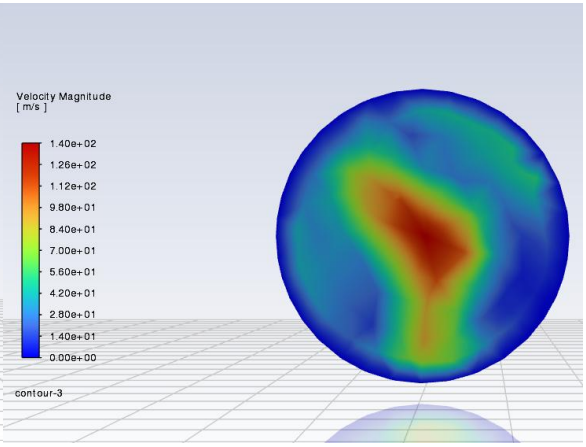
Model 2 Velocity Magnitude

8.3.4 Inside the Combustion Chamber

Model 1 shows a more centrally located velocity peak, indicating the flow has partically re-centralized after the swirl but in model 2, there is a visible asymmetric high velocity lobe on one side.



Model 1 Velocity Magnitude



Model 2 Velocity Magnitude

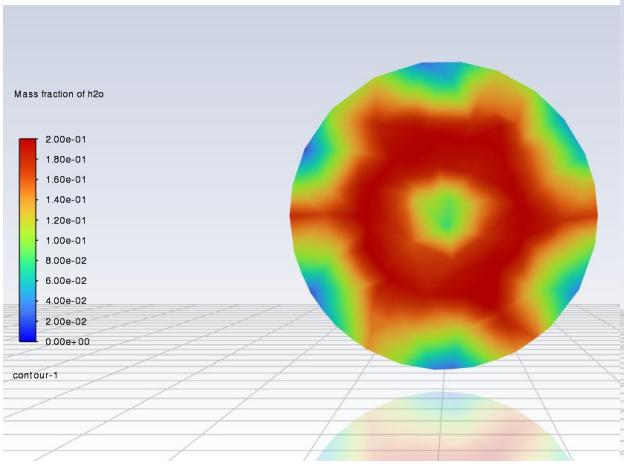
8.3.5 Interpretation

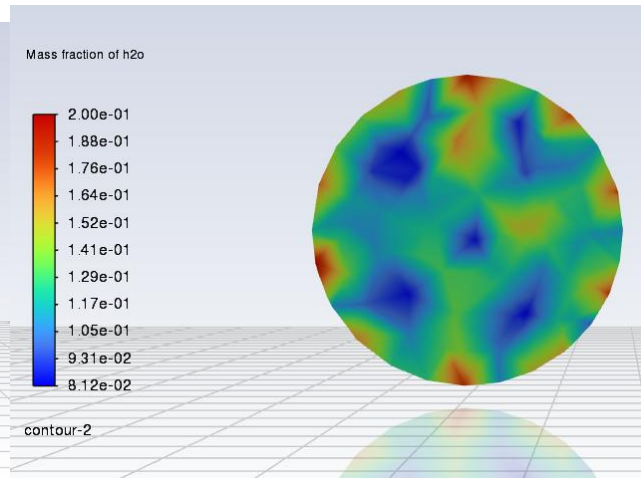
The velocity field in model 1 clearly showed presence of central recirculation zone (CRZ) a hallmark of swirl type injectors, which are considered to be better for the mixing of air and fuel whereas the velocity fields of model 2 was stagnant and displayed randomly distributed pockets of air fuel mixture

8.4 H_2O Mass Fraction

8.4.1 Inside the Injector

Model 1 shows a concentric annular H_2O ring like distribution with a maximum of 0.2 and lowest at the center and walls. This is due to the swirl induced and the swirl flame annular combustion zone formed due to it. It can also be seen in the temperature field in the same plane. Model 1 shows mass fractions ranging between 0.081 to 0.2 with a highly scattered irregular distribution, forming pockets of high and low concentration of the mixture.

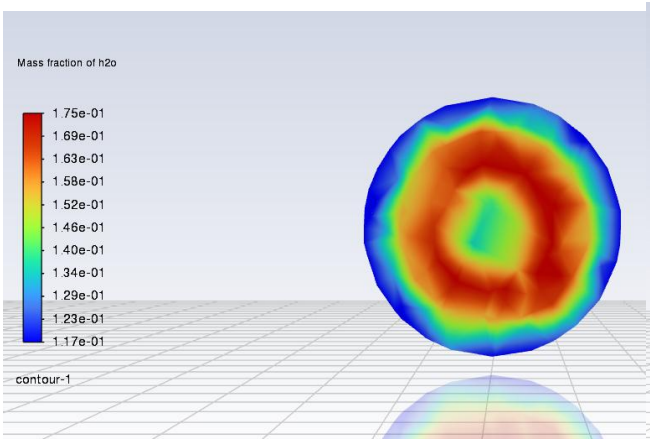


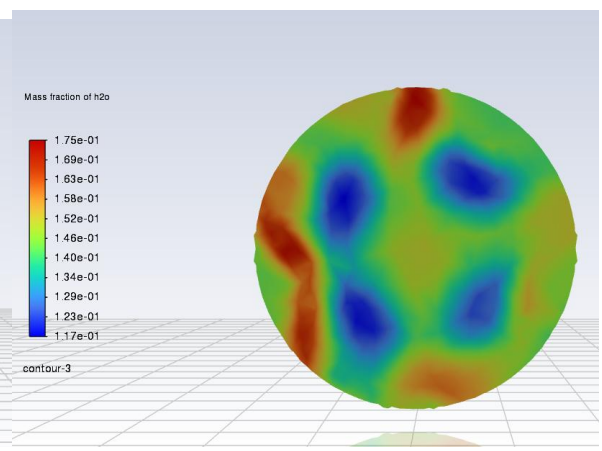


Mass Fraction of H_2O of Model 1 Mass Fraction of H_2O of Model 2

8.4.2 Injector Exit Plane

Model 1 shows a clear bull's eye pattern with a peak of 0.175 as an annular ring and diminishing towards the center and the edge. The jets visible in the beginning have merged into a coherent flame front. Model 2 shows the same max value at multiple discrete pockets that directly correspond to the individual jets from before.

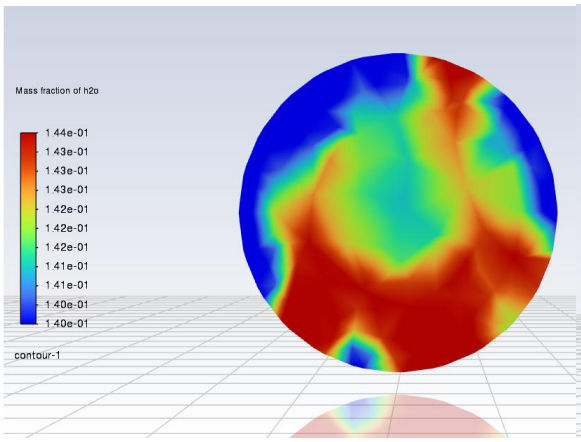


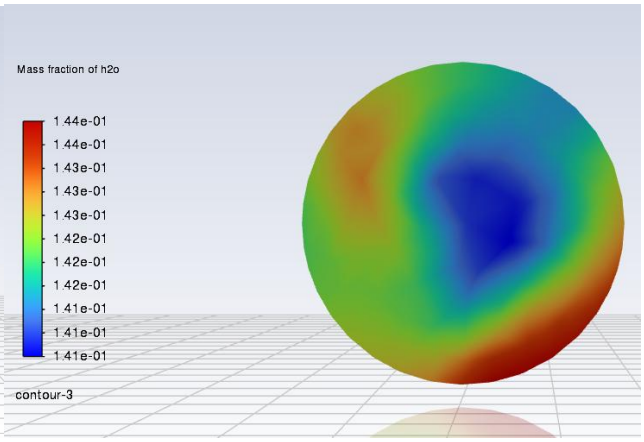


Mass Fraction of H_2O of Model 1 Mass Fraction of H_2O of Model 2

8.4.3 Inside the Combustion Chamber

Model 1 ranges from 0.140 to 0.144, similarly narrow but with a broader non uniform region visible which is still found to be converging to equilibrium. On the other hand, model 2 has a narrow range which is essentially fully mixed and equilibrated, which is due to the longer injector and combustion chamber.





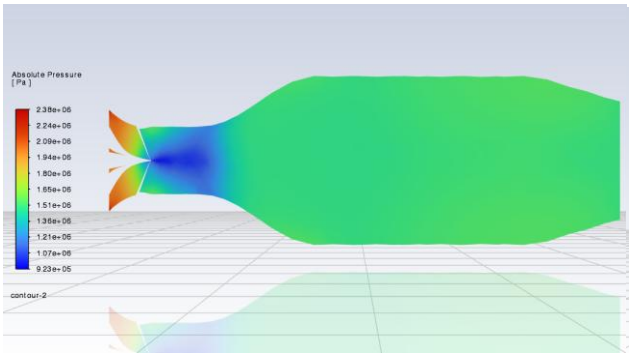
Mass Fraction of H_2O of Model 1 Mass Fraction of H_2O of Model 2

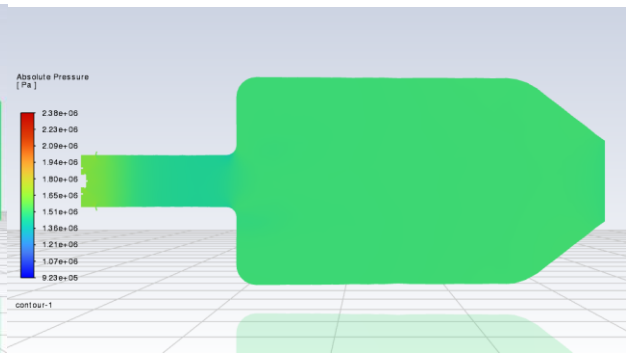
8.4.4 Interpretation

Both models achieve comparable H_2O levels, confirming similar overall combustion efficiency, but model 2 produces them via a swirl driven mechanism and under a very short injector and combustion chamber length.

8.5 Pressure

Model 1has 30% pressure loss compared to the 8% pressure loss that is found in model 2. It was calculated from the absolute pressure contour and the area weighted average of the inlet and outlet absolute pressures.





Absolute Pressure Model 1 Absolute Pressure Model 2

Model 1 has high pressure loss which is predictable and direct due to its design philosophy. It is due to sudden and severe area expansion from the narrow injector assembly into the large combustion chamber with the fan like spreading of swirling jets. It creates a large and irreversible loss due to expansion. The Borda-Carnot loss states that the pressure loss is directly proportional to the square of loss in velocity. In our case, the velocity is reducing from around 400-600m/s to 100-150 m/s, which is very large.

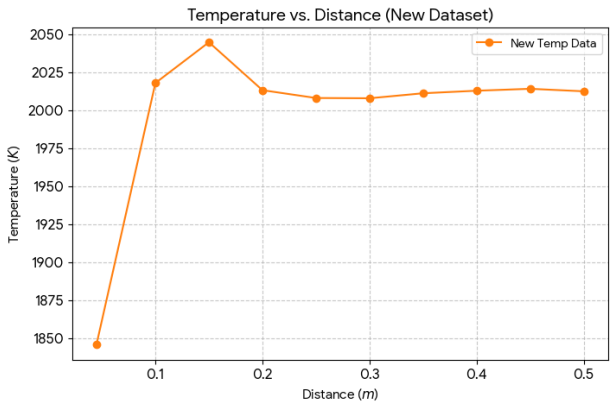
The high swirl number also generates strong tangential velocity components that do not contribute to axial flow progress. This energy is dissipated into heat through viscous shear rather than being retained as pressure inside the chamber.

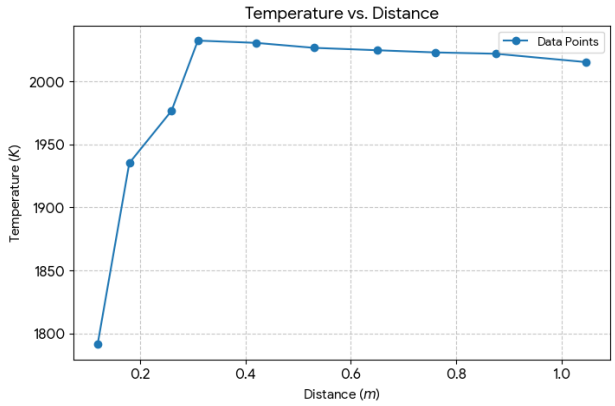
Model 2 retains its pressure due to lack of swirl intensity. However beneficial swirl may be, it is detrimental to the pressure inside the injector and the combustion chamber. Model 2 also operates under lesser static pressure. The flow is less energetically loaded at the inlet, so there is less energy to be lost as a whole. Since the flow of the injector develops more gradually, it reduces pressure loss inherently, using the longed axial distance to its advantage.

8.6 Temperature Across the Model

Model 1 starts at 1850K at the start of the injector, spikes to a peak of 2045K at 0.15m, then dropping back to 2010K to 2015K, stabilizing thereafter. Model 2 shows a similar pattern, starting at 1795K, peaking at 2030K at 0.3m, then gradually plateauing to 2015K at the outlet plane.

Model 1 reaches its peak temperature much earlier at 0.15m when compared to the 0.3m in model 2 and achieves a slightly higher peak at 2045K, whereas model 2 achieves a lower temperature at 2030K. This means that the flame is more compact and ignites faster in model 1. However, it shows a dip before restabilizing, suggesting a brief over rich zone followed by dilution. Model 1 shows a rise that is more gradual, and the plateau is smoother and longer indicating a more progressively developing flame. This can be since model 2 was longer than model 1 to account for mixing which happened only due to diffusivity and the turbulence driven mixing was low or close to none. Both models converge to a similar equilibrium temperature in the far field, confirming comparable overall heat release.



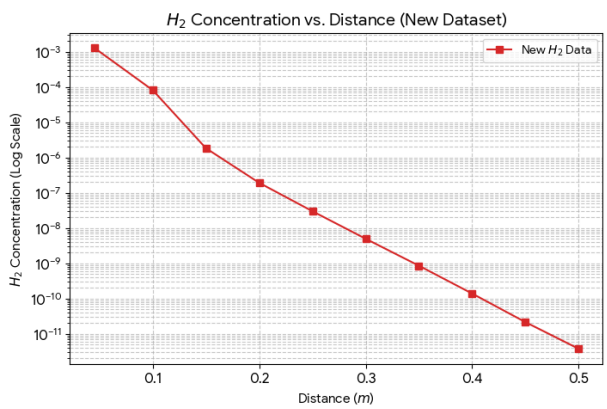


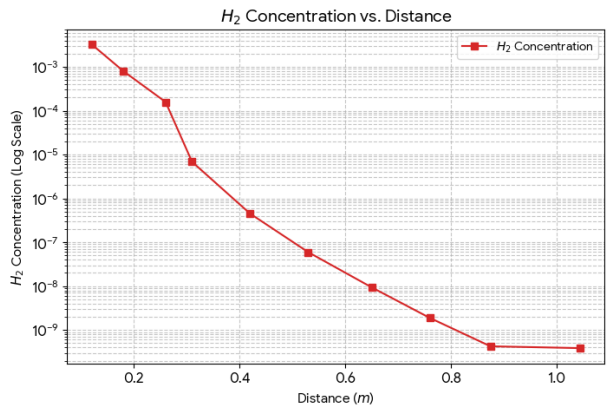
Distance vs Temperature (Model 1) Distance vs Temperature (Model 2)

8.7 H2 Concentration Across the Model

Model 1 starts at 10^{-3} at 0.05m and decays to $5x10^{-12}$ by 0.5m whereas in model 2 it starts at $3x10^{-3}$ at 0.15m and decays steadily on a log scale to 5x 10^{-10} by 0.9 to 1.05m. This is smooth and consistent over the full domain, reaching good combustion till the end.

Model 1 consumes hydrogen approximately 2 to 3 orders of magnitude more completely than model 2 despite having a much shorter axial distance. This better consumption indicates that the fuel and air mix well and the mixing is due to swirlers present in that model.



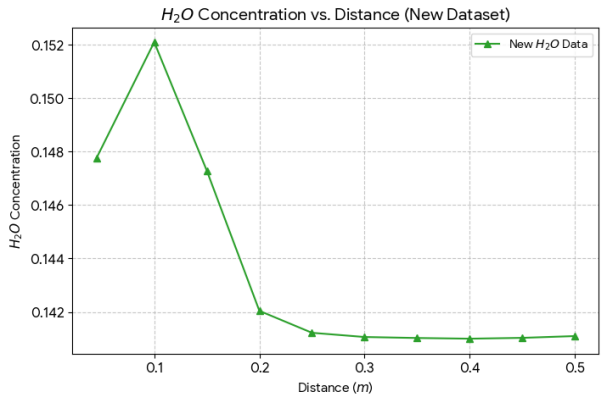


Distance vs H_2 (Model 1) Distance vs H_2 (Model 2)

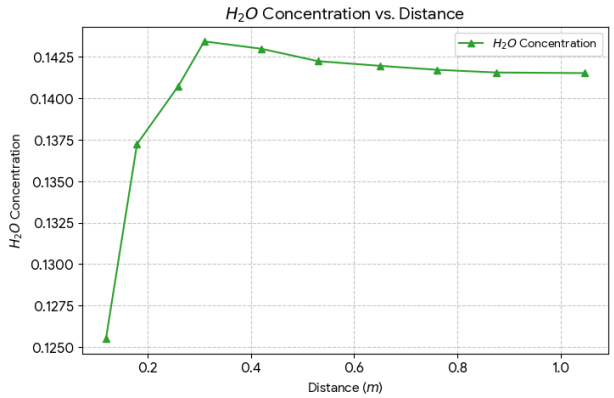
8.8 H2OConcentration Across the Model

Model 1 starts at 0.1478 at 0.05m, which is already high, meaning the early presence of combustion products. This peaks sharply at 0.152 at 0.1m, then drops and plateaus at around 0.1412 from 0.25m onwards. Model 2 starts at a very low 0.1255 at 0.15m and rises steeply to a peak of 0.1435 at 0.4m then gradually decreases and stabilizes at around 0.141 from 0.5m to 1.05m.

Model 1 produces more H_2O and at a higher initial peak which is consistent with its faster and more intense early combustion. The steep drop in that model is notable as it suggests that the initial rich combustion zone in the injector which then gets diluted as cooler air mixes in further downstream. Both the models converge to almost the same equilibrium value as they both have the same inlet mass flow rate and almost maximum combustion. This difference that would be observed between the models may stem because of incomplete mixing and combustion thereafter. As a whole, model 1 achieves the same level of H_2O production since it achieves faster than model 2 and does it more smoothly and is considered to be superior.



Distance vs H_2O (Model 1)



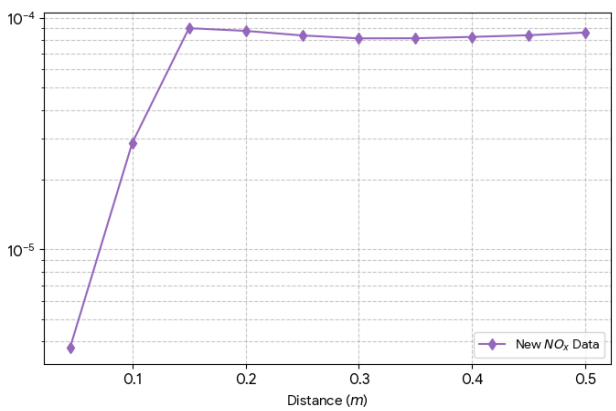
Distance vs H_2O (Model 2)

NO_x

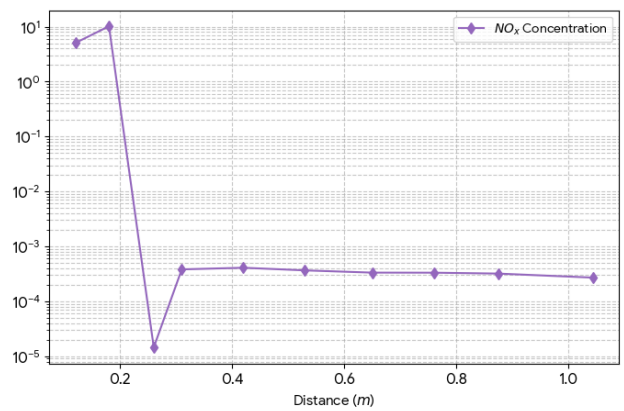
9.1 NO_x Tendency

Model 1 shows a clean, monotonically increasing profile that levels off quickly. Inside the injector it starts off at almost zero (3×10^{-6} Mole fraction of pollutant NO_x value) at 0.045m and it steadily rises to 10^{-4} PPM by 0.15m and then plateaus at around 8.5×10^{-5} Mole fraction of pollutant NOx value which is around 85 PPM from 0.2m to 0.5m. There is no spike in the profile and its smooth and asymptotic, which indicates gradual and controlled NO_x formation that reaches equilibrium early and does not increase by much downstream. Whereas model 2 exhibits a highly non monotonic NO_x profile along the axial distance. It starts at approximately 5000PPM in the injector at 0.15m and it spikes to 10000PPM at around 0.2m then continues to drop and level off at around 270 PPM

In terms of outlet NO_x value, model 1 is roughly 3 times lower at 87 PPM at the outlet compared to model 2 which has is around 270 at the outlet.



Distance vs NO_x (Model 1)



Distance vs NO_x (Model 2)

9.2 Flame and Temperature Structure

The NO_x tendencies are directly explained by the flame and temperature structures observed in the contour plots.

Model 1 produces a distributed and swirl stabilized flame where the heat of reaction is spread over a much larger volume and it is at a lower local intensity aswell. The temperature at the injector showed a smooth concentric annular pattern and has no extreme localized hot spots, with evident temperature distributed from 1800K to 2300K. It is because the temperature is distributed than localized. The smooth rise and early plateau in NO_x value reflects that NO_x forms at a steady and low rate as the flame establishes itself and finally stops increasing. It does as the fuel is consumed, and the

temperature also stabilizes. The absence of any sudden spikes confirms that no dangerous localized thermal hot spots exist in the flame structure.

In model 2, we can see a compact and high intensity flame anchored tightly near the injector exit. The heat of reactions contour shows a localized hot spot reaching up to 6200 W and the temperature zone is non uniform, and patchy fields are visible with those zones exceeding 2200K. It is precisely these spots and pockets that drive the production of NO_x up and it is due to the Zeldovich mechanism, which happens at high temperatures where oxygen, nitrogen are involved, production of NO_x takes place which is not desirable for the environment and the engine. This spike in NO_x can be seen at 0.2m is due to the hot spots in that cross section, and the subsequent collapse to around 270 PPM and stabilization occurs because the hot gasses rapidly mix and dilutes with the cooler surrounding, quenching the NO_x production.

In summary, model 1 has low and smooth NO_x since it has a much-distributed heat release and uniform temperature. Model 2 has high heat release and non-uniform temperature distribution leads to a NO_x spike and consequently a higher production.

9.3 Caution on Interpretation

There are several important caveats that needs to be acknowledged as a part of the simulation of the models.

9.3.1 Turbulence Chemistry Interaction Model

The models both use species transport with an eddy dissipation concept (EDC) combustion model. NO_x is usually a post processed pollutant using simplified thermal NO_x sub-models. The main mechanism is the Zeldovich thermal NO_x mechanism and is sensitive to the local temperature field. If the turbulence model slightly over or under predicts the temperatures, NO_x values can shift by a lot. The spike in model 2 could be an example of such mechanism and must be interpreted with care.

9.3.2 Grid Resolution Sensitivity

NO_x formation is highly sensitive to local temperature peaks which in turn depends on the mesh resolution in the reaction zone. If the mesh of model 2 was a bit coarser, it would probably smear out the peak temperature and the local NO_x formation be underestimated. As the other side to the coin, if the mesh is coarse, it may amplify the production of NO_x production.

9.3.3 NO_x Production

At the temperatures of 2000K to 2300K, thermal NO_x dominates and we are working only with thermal NO_x. But at leaner conditions such as the models we are working with, prompt NO_x and N_2O pathways can contribute meaningfully. This can be a factor for

model 1, where the flame is more distributed and potentially locally leaner may be underestimated, since it has better mixing and operates at a lower temperature and can benefit from other NO_x models as well.

RESULTS AND DISCUSSION

10.1 Engineering Comparison

The two hydrogen fuel injector models represent fundementally different combustion philosophies, and the CFD results show clear and consistent differences across the parameters and achieved similar results.

Model 1 employs a swirl-based stabilized injector configuration, where the hydrogen is injected at wider angles and interacts to form a coherent rotating flow structure. This generates a central recirculation zone (CRZ). It is visible in the temperature contours as well as the velocity contours. This continuously recirculates the hot combustion products back towards the injector's face. Even though it is being recirculated and the products undergo combustion, the NO_x value is less than the other model because the temperature of the combustion of the mixture is lesser or almost the temperature at which NO_x is produced. This recirculation acts as a self-sustaining ignition source, ensuring stable combustion across a wide operating range. It produces great uniformity since it is distributed volumetrically across the injector. Hydrogen burnout happens in a smaller space (within 0.5m) and NO_x formation is smooth, low and spike free, throughout the domain.

Model 2 also has a multi hole injector arrangement which is at angles alternating one after the other along the wall of the injector. It depends on diffusivity to mix the fuel and combust the mixture. But, as the hydrogen is radially injected into the axially moving air, it forms a collection of separate diffusion style reaction zones that only gradually merge as the flow develops downstream. It thus results in a compact and concentrated flame right near the exit of the injector. The flame is energetically efficient in the sense that it releases heat rapidly but does not do it evenly across the plane. When the flame is expanded and the mixing is essentially complete in the combustion chamber, it occurs over a longer axial distance than model 1.

Both models operate within the same conditions and the similar combustion chamber geometry and achieve comparable far field equilibrium temperatures and H_2O mass fractions, confirming that the overall combustion efficiency is similar. Even though they are comparable, they are achieved entirely differently. The spatial structure, rate and uniformity of the process and it has direct engineering consequences for emissions, hardware durability and operational flexibility which is explicitly displayed in the NO_x value at the outlet, despite the model 2 being more than twice the length axially than model 1. However if pressure loss is the only concern and it needs to be as low as it can be, model 2 would be recommended at this point in time, even with a little bit of work, model 1 looks more promising and has more potential to be worked on than model 2.

10.2 Compromises

No combustor design is perfect and without any compromises. Both models present genuine trade-offs, and they must be acknowledged correctly.

Model 1's advantages are clear in emissions, combustion completeness, and near field uniformity. The swirl stabilized flame is inherently more stable as it reintroduces the combustion products, and that is more robust to the small perturbations in the fuel air ratio, and the credit goes to CRZ. The lack of a spike in NO_x and 3 times lower NO_x value, are a great positive in this model since these engines are subject to emission regulations. The shorter injector and combustion chamber length is also a great advantage as it denotes that a compacter injector design is achievable. The swirl design may be subject to thermoacoustic instabilities as the rotating flow structure can couple with acoustic modes of the combustion chamber, thereby producing pressure oscillations that have potential to damage the hardware. This was not assessed in the current simulation and represents the most important uncertainty in evaluating this model. The pressure loss is also quite significant in model 1 as it loses almost 30% of its initial absolute pressure compared to the 8% lost by model 2.

Model 2's advantages lie in its simplicity and potentially stronger flame anchoring at lower load conditions. The high-momentum jets create strong local mixing that could resist blowouts at reduced fuel flow. The smooth and long temperature plateau is observed in the axial profile, and it is favorable turbine inlet temperature uniformity perspective, but this comes at a significant cost. It has localized hot spots that reach above 2200K in the injector, creating both NO_x penalty and a thermal fatigue risk for the liner walls. It also forms patchy hot and cold spots and the temperature is not equally spread out in certain places and even if it is able to be contained inside the injector, even a small shift in the operating conditions can cause catastrophic damage to the injector and the combustion chamber, thus having a risk of reliability.

10.3 Recommended Concept

Based on all the evidence from the CFD results, the obtained contour plots, axial line graphs and NO_x analysis, model 1 is the recommended concept out of the two and has better prospects for future development.

It is chosen since hydrogen combustion, the elimination of carbon-based pollutants inherent, which makes production of NO_x the main concern. Model 2 has 3 times less outlets of NO_x and complete absence of NO_x spikes. It also boasts almost complete consumption of the hydrogen injected for combustion within the shorter axial length compared to twice the axial length in model 2. Thus, it is decisively superior from

engineering and environmental standpoints. The flame in model 1 is also well established due to the formation of central recirculation zones, which is also established in gas turbine literature as the preferred approach for low emission combustor design. It also means there are a substantial number of tools to aid in the further development of such a swirl induced injector. The flame also reduces thermal loading on the injector, which directly translates to longer component life and reliability.

10.4 Future Work

There are a lot of steps and work that is required before either design could progress towards physical testing or detailed engineering design.

The most immediate need is a mesh independence study focused specifically on the injector zone, where both temperature gradients and NO_x formation rates are the steepest. A similar pattern of increase in NO_x can be found in both the models while it being a huge variation in model 2. Focusing on mesh may resolve the peak temperature, H_2 burn out length and NO_x values.

The major pressure lost must be addressed as the industry standard is 5% to 7% and model 1 looses almost 30% of its initial absolute thrust which is nowhere close to being acceptable in the industry. It can be done by a lot of ways, some of them are, adding a diffuser transition zone at the injector exit, increasing the injector tube diameter by a bit, adding guide vanes at the chamber entrance, etc. By adopting 1 or more of these techniques, a pressure loss of 5% is possible for model 1 which has a swirl type injector.

Following it, transient simulation using LES (Large Eddy Simulation) or RANS should be performed for model 1 as it can help assess the thermoacoustic stability. This is particularly important for swirl stabilized flames which are known to be susceptible to combustion instabilities.

A parametric study on the number of swirlers and the angle it is swirling at, and the number of fuel injector holes and the angle it is injected in is required to optimize the mixing of the fuel and air to its maximum potential.

Finally, part load and ignition transient simulations should be conducted to assess the behavior outside the single design point condition examined in this report. A combustor must operate reliably across a wide turndown range, perform well at its full load and be reliable at the same time.

CONCLUSION

11.1 Technical Conclusions

This study compares two hydrogen fuel injector designs using 3D reacting flow CFD simulations in Ansys Fluent with species transport enabled.

It was found that model 1, with the swirl stabilized injector, outperforms model 2 on every important performance metric, which is in a modern hydrogen combustor. It achieves faster flame establishment in a more compact design and has significantly lower NO_x emissions. Its distributed, swirl driven flame avoids the localized thermal hot spots that drive exponential thermal NO_x formation, which can be seen in model 2.

Both the models achieve similar overall combustion completeness in teh far field combustion chamber despite the difference in their axial lengths, with nearly identical H_2O equilibrium at the end with similar overall temperatures. This confirms that the difference between the models is due to the temporal organization of that release, and it's something that has profound implications for emissions, hardware life, reliability, and operability.

11.2 Knowledge Gained

The most important lesson from this comparison is that the flame structure governs emissions far more than total heat release. Both the models burn the same amount of fuel and oxygen, producing the same products at equilibrium, yet the NO_x output differs by 3 times simply because of how spatially organized the combustion is.

The mass fraction alone is an insufficient indicator of combustion quality. When examining the residual H_2 concentration at the outlet, and the spatial structure of the temperature field, the superiority of model 1 becomes apparent.

The last lesson is the value and limitations of steady state RANS simulation. The results produced here are physically coherent, internally consistent and sufficient to distinguish between the two designs with confidence. However, steady RANS cannot capture flame dynamics, thermoacoustic coupling, or ignition transients all of which are critical to a real combustor. Even though model 1 may be a good model in this report, deep analysis is required before it can be considered to be adapted in a physical real-world test.

APPENDIX

The models submitted in Assignment 1,2 and 3 were changed to the models compared now, since the models used in the previous assignments were similar and produced almost the same results and didn't not leave much room for improvement. It was because both the models relied on diffusion for the mixing of fuel and air. Further research into swirl type injectors made it look promising, and the ability to continue the research further could make more sense, since it is already a critical design in the current injector sector for hydrogen engines. More over the diffusion models created for the assignments varied by more than 10 times to the newly created swirl type injector and were not comparable, so the positive points from those injectors used in the assignments were taken and a new injector was designed, despite being longer had the same concept of the injector it was replacing.

As a result, the models used for this final report were the new model 1 (swirl type injector) which has the best results we could have expected, along with model 2 outperforming its predecessors and almost keeping up with model 1.